A MONOMORPHISM OF ALGEBRAIC STACKS IS SEPARATED AND DM, AND SO IS AN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The statement that the diagonal of a monomorphism of algebraic stacks is an isomorphism. Invoked at the start of the counted proof (source lines 1165-1167) as a cross-chapter pointer. Reduces separatedness and DM-ness to properties of an isomorphism of algebraic spaces.
- The statement that an immersion of algebraic stacks is a monomorphism. Invoked at the end of the counted proof (source lines 1171-1173) as a cross-chapter pointer. Transfers the first assertion to immersions.

Source reference. Original reference stacks-properties-lemma-immersion-monomorphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1172–1172. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-immersion-monomorphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-monomorphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1167–1167. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-monomorphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *A monomorphism of algebraic stacks is separated and DM. The same is true for immersions of algebraic stacks.*

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. If $f : \mathcal{X} \to \mathcal{Y}$ is a monomorphism of algebraic stacks, then Δ_f is an isomorphism, see Properties of Stacks, Lemma stacks-properties-lemma-monomorphism. Since an isomorphism of algebraic spaces is proper and unramified we see that f is separated and DM. The second assertion follows from the first as an immersion is a monomorphism, see Properties of Stacks, Lemma stacks-properties-lemma-immersion-monomorphism. □